Gröbner Basis Methods for the Kinematics of Planar and Spatial Robotic Mechanisms

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by

Ayush Verma

23MF10008

Under the supervision of

Professor Mousumi Mandal

Department of Mathematics

Indian Institute of Technology Kharagpur

Kharagpur – 721302, India



Autumn Semester, 2026–27

12th November, 2026

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(Ayush Verma)
(23MF10008)

DEPARTMENT OF MATHEMATICS
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
KHARAGPUR – 721302, INDIA

CERTIFICATE

This is to certify that the project report entitled **"Gröbner Basis Methods for the Kinematics of Planar and Spatial Robotic Mechanisms"** submitted by **Ayush Verma** (Roll No. 23MF10008) to Indian Institute of Technology Kharagpur towards partial fulfilment of the requirements for the award of the degree of Bachelor of Technology (Honours) in Manufacturing Science and Engineering is a record of work carried out by him under my supervision and guidance during Autumn Semester, 2026–27.

Date: 12th November, 2026
Place: Kharagpur

Professor Mousumi Mandal
Department of Mathematics
Indian Institute of Technology Kharagpur
Kharagpur – 721302, India

Abstract

The kinematics of a robot is a system of polynomial equations once every joint angle θ is replaced by the pair $(\cos\theta, \sin\theta)$ and every rigid-body or leg constraint is written as a squared distance. Solving the system is then a question about an affine variety, and Gröbner bases give an algorithmic answer. This report first sets out the required theory (polynomial ideals, monomial orderings, the division algorithm, Gröbner bases, and Buchberger's algorithm and criteria), stating definitions, algorithms and theorems without proofs. The theory is then applied to three example test instances used in this work: the inverse kinematics of a planar 2R1P serial robot, the forward kinematics of a planar 3-RPR parallel manipulator, and the forward kinematics of an octahedral (3-3) Stewart–Gough platform. For each, the geometry is converted into polynomials $f_1, \ldots, f_n$ with an explicit variable ordering, a lexicographic Gröbner basis is used to reduce the system to triangular form, and the real, physically admissible solutions are extracted: two valid configurations for the 2R1P robot, a single real pose for the 3-RPR manipulator, and four real poses (two mirror pairs) out of sixteen complex ones for the octahedral platform. Finally, the execution time of a SymPy-based Python implementation of Buchberger's algorithm is compared with Macaulay2 under the `lex`, `grlex` and `grevlex` orderings. Macaulay2 is faster by three to four orders of magnitude in nearly all cases, and `lex` is the most expensive ordering in both systems.

Keywords: Gröbner bases, Buchberger's algorithm, robot kinematics, 3-RPR manipulator, Stewart–Gough platform, Macaulay2, SymPy.

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List of Symbols

$\mathbb{Q},\ \mathbb{R},\ \mathbb{C}$	Fields of rational, real and complex numbers
$k[x_1, \ldots, x_n]$	Polynomial ring over a field k
x^α	Monomial $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, $\alpha \in \mathbb{Z}_{\geq 0}^n$
$\langle f_1, \ldots, f_s \rangle$	Ideal generated by $f_1, \ldots, f_s$
$\mathbf{V}(f_1, \ldots, f_s)$	Affine variety (common zero set) of $f_1, \ldots, f_s$
$>_{\text{lex}},\ >_{\text{grlex}},\ >_{\text{grevlex}}$	Lexicographic, graded lexicographic, graded reverse lexicographic orders
$\text{LM}(f),\ \text{LC}(f),\ \text{LT}(f)$	Leading monomial, leading coefficient, leading term of f
$\overline{f}^F$	Remainder of f on division by the ordered tuple F
$S(f, g)$	S-polynomial of f and g
G	Gröbner basis of an ideal
$f_1, \ldots, f_n$	Polynomial system obtained from a mechanism
$\theta_1, \theta_2, \theta$	Joint angles and end-effector orientation of the 2R1P robot
$c_i,\ s_i$	$\cos\theta_i$ and $\sin\theta_i$
$L_1,\ L_2,\ p$	Link lengths and prismatic extension of the 2R1P robot
(a, b)	End-effector position of the 2R1P robot
(x, y, ϕ)	Pose of the 3-RPR moving platform; $c = \cos\phi$, $s = \sin\phi$
$A_i,\ B_i$	Base joints and platform joints (3-RPR); base joints and platform nodes B_i, P_i (Stewart)
d_i	Leg length of the 3-RPR manipulator
$P_i = (x_i, y_i, z_i)$	Moving-platform spherical nodes of the Stewart–Gough platform
l_k	Leg length (squared leg length l_k^2) of the Stewart–Gough platform

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1 Introduction

A robot is described by joint variables, and the position and orientation of its end effector are functions of them. Two questions dominate: *forward kinematics* (given the joints, find the pose) and *inverse kinematics* (given the pose, find the joints). For serial robots the inverse problem is the hard one; for parallel robots it is the forward problem. In both cases the unknowns satisfy equations that become *polynomial* once trigonometric functions are replaced by cosine–sine pairs and distances are written as squares. Finding all solutions of such a system is a problem in computational algebraic geometry, and Gröbner bases are its standard tool [1].

This report has two parts. Part I (Section 2) states the required theory: ideals, varieties, monomial orderings, the division algorithm, Gröbner bases, and Buchberger's algorithm and criteria. Only definitions, algorithms and theorem statements are given; proofs are in [1]. Part II (Sections 3–5) treats three mechanisms with the same four-part template: problem statement and diagram, conversion of the kinematics to a polynomial system $f_1, \dots, f_n$, the Gröbner basis and the real solutions, and a timing comparison of Python and Macaulay2. All numerical parameters are *example test instances used in this work.*

- **Planar 2R1P robot** (inverse kinematics): two revolute joints and one prismatic joint; six equations in five unknowns.
- **Planar 3-RPR parallel manipulator** (forward kinematics): four equations in four unknowns, with a classical bound of six assembly modes [2, 3].
- **Octahedral (3-3) Stewart–Gough platform** (forward kinematics): nine equations in nine unknowns and sixteen complex solutions [4].

The *Python* computations use a SymPy-based implementation of Buchberger's algorithm with hand-written modifications that expose and simulate its individual steps. The *Macaulay2* computations use the built-in `gb` command. The three orderings `lex`, `grlex` and `grevlex` are compared; Macaulay2 times are those of `gb` alone, Python times are those of the whole script run.

2 Theoretical Foundations (Part I)

Throughout, k is a field ($\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$) and $k[x_1, \dots, x_n]$ is the ring of polynomials in $x_1, \dots, x_n$. This section follows [1]; proofs are omitted.

2.1 Polynomial rings, ideals and affine varieties

Definition 2.1 (Ideal)**.** A subset $I \subseteq k[x_1, \dots, x_n]$ is an *ideal* if $0 \in I$, $f + g \in I$ whenever $f, g \in I$, and $hf \in I$ whenever $f \in I$ and $h \in k[x_1, \dots, x_n]$. For $f_1, \dots, f_s$ the set $\langle f_1, \dots, f_s \rangle = \{h_1 f_1 + \cdots + h_s f_s : h_i \in k[x_1, \dots, x_n]\}$ is the ideal they *generate*.

Definition 2.2 (Affine variety)**.** The *affine variety* of $f_1, \dots, f_s$ is $\mathbf{V}(f_1, \dots, f_s) = \{a \in k^n : f_i(a) = 0 \text{ for all } i\}$.

Proposition 2.3. *If $I = \langle f_1, \dots, f_s \rangle$ then $\mathbf{V}(I) = \mathbf{V}(f_1, \dots, f_s)$. Hence the variety depends only on the ideal, not on the chosen generators.*

This is the key freedom exploited below: a system can be replaced by another generating set of the same ideal with the same solutions. Over $k = \mathbb{C}$, $\mathbf{V}(I) = \emptyset$ if and only if $1 \in I$ (Weak Nullstellensatz). The *ideal membership problem* asks whether a given f lies in I.

2.2 Monomial orderings and the division algorithm

Definition 2.4 (Monomial ordering)**.** A *monomial ordering* on $k[x_1, \ldots, x_n]$ is a relation $>$ on $\mathbb{Z}^n_{\geq 0}$ (the exponents) that is a total ordering, satisfies $\alpha > \beta \Rightarrow \alpha + \gamma > \beta + \gamma$, and is a well-ordering.

Let $\alpha, \beta \in \mathbb{Z}^n_{\geq 0}$ and $|\alpha| = \alpha_1 + \cdots + \alpha_n$, with $x_1 > x_2 > \cdots > x_n$.

- **Lexicographic (`lex`):** $\alpha >_{\text{lex}} \beta$ if the leftmost nonzero entry of $\alpha - \beta$ is positive.
- **Graded lexicographic (`grlex`):** $\alpha >_{\text{grlex}} \beta$ if $|\alpha| > |\beta|$, or $|\alpha| = |\beta|$ and $\alpha >_{\text{lex}} \beta$.
- **Graded reverse lexicographic (`grevlex`):** $\alpha >_{\text{grevlex}} \beta$ if $|\alpha| > |\beta|$, or $|\alpha| = |\beta|$ and the *rightmost* nonzero entry of $\alpha - \beta$ is negative.

For example, with $x > y > z$, $x^2z > xy^2$ in `lex` and `grlex`, but $xy^2 > x^2z$ in `grevlex`. For $f = \sum a_\alpha x^\alpha \neq 0$ the *multidegree* is $\max\{\alpha : a_\alpha \neq 0\}$, the *leading coefficient* is $\mathrm{LC}(f) = a_{\mathrm{multideg}(f)}$, the *leading monomial* is $\mathrm{LM}(f) = x^{\mathrm{multideg}(f)}$ and the *leading term* is $\mathrm{LT}(f) = \mathrm{LC}(f)\,\mathrm{LM}(f)$.

Theorem 2.5 (Division algorithm)**.** *Fix a monomial ordering and an ordered tuple $F = (f_1, \ldots, f_s)$ of nonzero polynomials. Every f can be written $f = a_1 f_1 + \cdots + a_s f_s + r$, where $r = 0$ or no term of r is divisible by any $\mathrm{LT}(f_i)$, and $\mathrm{multideg}(a_i f_i) \leq \mathrm{multideg}(f)$ whenever $a_i f_i \neq 0$. The remainder is denoted $\overline{f}^{\,F}$.*

The algorithm repeatedly divides: if some $\mathrm{LT}(f_i)$ divides $\mathrm{LT}(p)$ of the current polynomial p, subtract $\frac{\mathrm{LT}(p)}{\mathrm{LT}(f_i)} f_i$; otherwise move $\mathrm{LT}(p)$ to the remainder. For a general F the remainder depends on the ordering of F and $r = 0$ is not equivalent to $f \in \langle F \rangle$. Gröbner bases remove this defect.

2.3 Monomial ideals, Hilbert's Basis Theorem and Gröbner bases

Definition 2.6 (Monomial ideal)**.** An ideal generated by monomials is a *monomial ideal.*

Theorem 2.7 (Dickson's Lemma)**.** *Every monomial ideal $I = \langle x^\alpha : \alpha \in A \rangle$ is generated by a finite subset of $\{x^\alpha : \alpha \in A\}$.*

Theorem 2.8 (Hilbert Basis Theorem)**.** *Every ideal $I \subseteq k[x_1, \ldots, x_n]$ is finitely generated.*

Fix a monomial ordering and set $\langle \mathrm{LT}(I) \rangle = \langle \mathrm{LT}(f) : f \in I \setminus \{0\} \rangle$.

Definition 2.9 (Gröbner basis)**.** A finite set $G = \{g_1, \ldots, g_t\} \subseteq I$ is a *Gröbner basis* of I if $\langle \mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_t) \rangle = \langle \mathrm{LT}(I) \rangle$.

Theorem 2.10. *Every nonzero ideal has a Gröbner basis, and every Gröbner basis of I generates I. If G is a Gröbner basis of I, then $\overline{f}^{\,G}$ is unique for every f (independent of the order of G), and $f \in I$ if and only if $\overline{f}^{\,G} = 0$.*

Definition 2.11 (Reduced Gröbner basis)**.** A Gröbner basis G is *reduced* if $\mathrm{LC}(g) = 1$ for all $g \in G$, and no monomial of g lies in $\langle \mathrm{LT}(G \setminus \{g\}) \rangle$.

Theorem 2.12. *Every nonzero ideal has a unique reduced Gröbner basis for a given monomial ordering.*

2.4 S-polynomials and Buchberger's algorithm

Definition 2.13 (S-polynomial)**.** For nonzero f, g let $x^\gamma = \mathrm{lcm}(\mathrm{LM}(f), \mathrm{LM}(g))$. Then

$$S(f,g) = \frac{x^\gamma}{\mathrm{LT}(f)}\, f - \frac{x^\gamma}{\mathrm{LT}(g)}\, g.$$

The S-polynomial cancels the leading terms of f and g and exposes what the leading-term ideal might be missing.

Theorem 2.14 (Buchberger's Criterion)**.** *A finite set $G = \{g_1, \dots, g_t\}$ generating I is a Gröbner basis of I if and only if $\overline{S(g_i, g_j)}^{G} = 0$ for all $i \neq j$.*

Algorithm 2.15 (Buchberger's Algorithm)**.** *Input:* $F = (f_1, \dots, f_s)$ and a monomial ordering. *Output:* a Gröbner basis G of $\langle F \rangle$.

1. $G := F$.
2. Repeat: set $G' := G$; for every pair $p \neq q$ in G' compute $r := \overline{S(p, q)}^{G'}$ and, if $r \neq 0$, add r to G. Stop when $G = G'$.
3. Return G (then minimise and reduce to obtain the reduced Gröbner basis).

The algorithm terminates by the Hilbert Basis Theorem. Most S-polynomials reduce to 0, so criteria to discard pairs are essential in practice.

Proposition 2.16 (First Criterion)**.** *If* $\mathrm{LM}(f)$ *and* $\mathrm{LM}(g)$ *are relatively prime, then* $\overline{S(f, g)}^{\{f,g\}} = 0$*, so the pair (f, g) can be skipped in Buchberger's algorithm.*

2.5 From kinematic linkages to polynomial systems

A mechanism is described by its *configuration space*: all values of joint variables and pose variables compatible with its rigid links. Three steps turn it into an ideal.

1. **Rational parametrisation.** A revolute joint angle θ is replaced by two variables (c, s) with $c^2 + s^2 = 1$; a prismatic joint contributes a variable p directly; the position of a link end is then polynomial in these.
2. **Constraints.** Loop closures and orientation conditions give polynomial equations; for parallel and spatial platforms each leg or rigid edge is a squared-distance condition $\|P - Q\|^2 = \ell^2$.
3. **Solving.** With variables ordered, a `lex` Gröbner basis reduces the system to a triangular one that is solved by back-substitution; only the points of $\mathbf{V}(I) \cap \mathbb{R}^n$ that satisfy physical limits (such as $0 \leq p \leq 1$) are admissible.

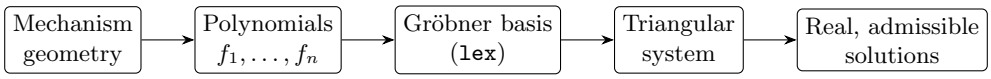


Figure 1: Workflow used for each case study.

3 Case Study 1: Planar 2R1P Robot (Inverse Kinematics)

3.1 Problem statement and diagram

Figure 2 shows a planar robot with two revolute joints R_1, R_2 and one prismatic joint. Link 1 is revolute at the origin, has length $L_1 = 2$ and joint angle θ_1. Link 2 starts at the second revolute joint R_2 with base length $L_2 = 1$ and prismatic extension p, $0 \le p \le 1$. The end-effector orientation is $\theta = \theta_1 + \theta_2$. For the example test instance used in this work, the target pose is

$$a = \tfrac{3}{2}, \qquad b = \tfrac{39}{20}, \qquad \theta = 0^\circ,$$

and the goal is to find θ_1, θ_2, p (inverse kinematics).

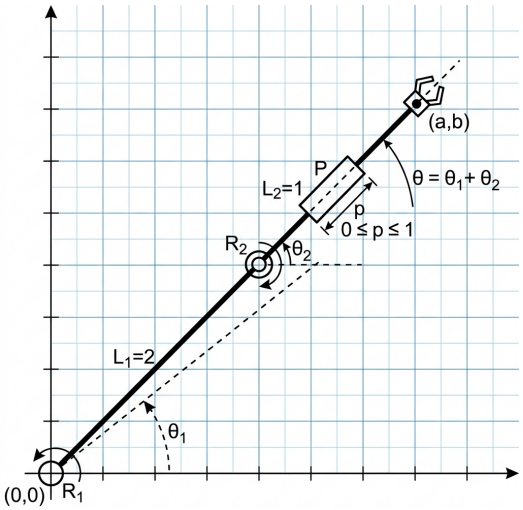


Figure 2: Planar 2R1P robot (schematic, not to scale).

3.2 Kinematic formulation to polynomial system

Let $c_1 = \cos\theta_1$, $s_1 = \sin\theta_1$, $c_2 = \cos\theta_2$, $s_2 = \sin\theta_2$. *Step 1 (position).* The end effector is at

$$a = 2c_1 + (1+p)\cos\theta, \qquad b = 2s_1 + (1+p)\sin\theta.$$

Step 2 (angle addition). Since $\theta = \theta_1 + \theta_2$,

$$\cos\theta = c_1c_2 - s_1s_2, \qquad \sin\theta = s_1c_2 + c_1s_2.$$

Step 3 (circle constraints). $c_1^2 + s_1^2 = 1$ and $c_2^2 + s_2^2 = 1$. *Step 4 (target pose).* With $\theta = 0^\circ$ we have $\cos\theta = 1$, $\sin\theta = 0$; substituting $a = \frac{3}{2}$, $b = \frac{39}{20}$ gives

$$2c_1 + p - \tfrac{1}{2} = 0, \qquad 2s_1 - \tfrac{39}{20} = 0,$$

and the orientation condition reads $c_1c_2 - s_1s_2 = 1$, $s_1c_2 + c_1s_2 = 0$. With the variable ordering $c_1 > s_1 > c_2 > s_2 > p$, the system is

$$\begin{aligned} f_1 &= c_1^2 + s_1^2 - 1, & f_2 &= c_2^2 + s_2^2 - 1, & f_3 &= c_1c_2 - s_1s_2 - 1, \\ f_4 &= c_1s_2 + c_2s_1, & f_5 &= 2c_1 + p - \tfrac{1}{2}, & f_6 &= 2s_1 - \tfrac{39}{20}. \end{aligned}$$

There are six equations in five unknowns; the system is overdetermined but consistent.

3.3 Algebraic solution

The reduced `lex` Gröbner basis of $\langle f_1, \dots, f_6 \rangle$ has five elements (verified by computer algebra):

$$c_1 + \tfrac{p}{2} - \tfrac{1}{4}, \quad s_1 - \tfrac{39}{40}, \quad c_2 + \tfrac{p}{2} - \tfrac{1}{4}, \quad s_2 + \tfrac{39}{40}, \quad p^2 - p + \tfrac{21}{400}.$$

It is triangular: the last element is univariate in p with roots $p = \frac{1}{2} \pm \frac{\sqrt{79}}{20}$, and the others fix $c_1 = c_2 = \frac{1}{4} - \frac{p}{2} = \mp\frac{\sqrt{79}}{40}$ and $s_1 = -s_2 = \frac{39}{40}$. Both real solutions satisfy $0 \le p \le 1$ and are therefore physical (Table 1).

	p	$c_1 = c_2$	$s_1 = -s_2$	θ_1	θ_2
Solution 1	$\frac{1}{2} - \frac{\sqrt{79}}{20} \approx 0.0556$	$\frac{\sqrt{79}}{40} \approx 0.2222$	$\frac{39}{40} = 0.9750$	77.16°	-77.16°
Solution 2	$\frac{1}{2} + \frac{\sqrt{79}}{20} \approx 0.9444$	$-\frac{\sqrt{79}}{40} \approx -0.2222$	$\frac{39}{40} = 0.9750$	102.84°	-102.84°

Table 1: Real solutions of the 2R1P inverse kinematics (both within $0 \le p \le 1$).

In both, $\theta_2 = -\theta_1$, so the end effector keeps orientation $\theta = 0^\circ$; the two configurations are related by $\theta_1 \mapsto 180^\circ - \theta_1$ (the arm leans left or right of the vertical).

3.4 Computational benchmarks

Table 2 lists the execution times. The Macaulay2 time is that of `gb`; the Python time is that of the whole run.

Ordering	Macaulay2 (ms)	Python (s)	Python / Macaulay2
`lex`	9.82	1.025	$\approx 1.0 \times 10^2$
`grlex`	0.499	0.931	$\approx 1.9 \times 10^3$
`grevlex`	0.452	1.015	$\approx 2.2 \times 10^3$

Table 2: Execution time for the 2R1P system.

In Macaulay2 `lex` is about twenty times slower than the graded orderings, which agrees with the general expectation that the lexicographic ordering is costlier. In Python all three orderings take about one second, which suggests that fixed overhead of the implementation dominates for a system this small.

4 Case Study 2: Planar 3-RPR Parallel Manipulator (Forward Kinematics)

4.1 Problem statement and diagram

The planar 3-RPR manipulator in Figure 3 has a fixed base and a moving platform joined by three legs, each a revolute–prismatic–revolute chain with an actuated prismatic joint. Gröbner bases have been used on its forward problem in [2] and on degenerate families in [3]. For the example test instance used in this work, the fixed-base revolute joints are $A_1 = (-2, 0)$, $A_2 = (2, 0)$, $A_3 = (0, 3)$; the platform joints, in the local platform frame, are $B_1 = (-1, 0)$, $B_2 = (1, 0)$, $B_3 = (0, 1)$; and the actuated leg lengths satisfy $d_1^2 = 2$, $d_2^2 = 2$, $d_3^2 = 1$. The goal is the platform pose (x, y, ϕ), with $c = \cos\phi$, $s = \sin\phi$ (forward kinematics).

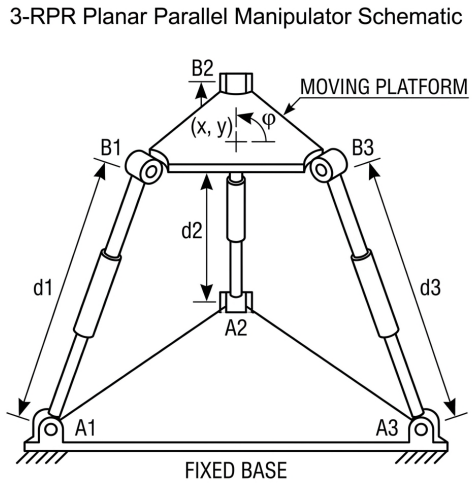


Figure 3: Planar 3-RPR parallel manipulator (schematic, not to scale).

4.2 Kinematic formulation to polynomial system

With $R(\phi) = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$, the world position of platform joint i is $(x, y)^T + R(\phi)B_i$, and leg i imposes

$$\left\| (x, y)^T + R(\phi)B_i - A_i \right\|^2 = d_i^2, \qquad c^2 + s^2 = 1.$$

Leg 1: $(x - c + 2)^2 + (y - s)^2 = 2$ expands to $x^2 + y^2 + 4x - 2xc - 4c - 2ys + (c^2 + s^2) + 2 = 0$; using $c^2 + s^2 = 1$ the constant becomes 3. *Leg 2:* $(x + c - 2)^2 + (y + s)^2 = 2$ gives constant 3 in the same way. *Leg 3:* $(x - s)^2 + (y + c - 3)^2 = 1$ gives constant 9. With variable ordering $x > y > c > s$:

$$\begin{aligned} f_1 &= x^2 + y^2 + 4x - 2xc - 4c - 2ys + 3, \\ f_2 &= x^2 + y^2 + 2xc - 4x - 4c + 2ys + 3, \\ f_3 &= x^2 + y^2 - 2xs + 2yc - 6y - 6c + 9, \\ f_4 &= c^2 + s^2 - 1. \end{aligned}$$

4.3 Algebraic solution

The reduced `lex` Gröbner basis has four elements (verified by computer algebra):

$$x = \tfrac{1}{7560}\left(25181s^5 + 14851s^3 + 7560s\right), \qquad y = 1 + \tfrac{1}{84}\left(169s^4 + 107s^2\right),$$
$$c = 1 - \tfrac{1}{2520}\left(169s^4 + 1199s^2\right),$$

$$0 = s^2\left(169s^4 + 263s^2 + 144\right) = s^2(13s^2 - 7s + 12)(13s^2 + 7s + 12).$$

The univariate polynomial in s has degree six, matching the classical bound of six assembly modes of a 3-RPR manipulator [2]. Both quadratic factors have negative discriminant $49 - 624 = -575$, so the only real root is $s = 0$ (a double root). Back-substitution gives $c = 1$, $x = 0$, $y = 1$. There is therefore a single real solution:

$$(x, y, c, s) = (0,\ 1,\ 1,\ 0), \qquad \phi = 0^\circ.$$

Direct check: with $\phi = 0^\circ$ the platform joints are at $(-1, 1)$, $(1, 1)$, $(0, 2)$ and their distances to A_1, A_2, A_3 are $\sqrt{2}$, $\sqrt{2}$, 1, as required; the platform is simply translated one unit upward.

4.4 Computational benchmarks

Ordering	Macaulay2 (ms)	Python (s)	Python / Macaulay2
`lex`	3.91	8.575	$\approx 2.2 \times 10^3$
`grlex`	0.583	4.37	$\approx 7.5 \times 10^3$
`grevlex`	0.638	3.826	$\approx 6.0 \times 10^3$

Table 3: Execution time for the 3-RPR system.

Both systems rank the orderings the same way: `lex` is slowest (about 6.7× slower than `grlex` in Macaulay2 and 2× slower than `grevlex` in Python), while `grlex` and `grevlex` are close. The gap between the two systems is three to four orders of magnitude.

5 Case Study 3: Octahedral (3-3) Stewart–Gough Platform

5.1 Problem statement and diagram

A Stewart–Gough platform connects a base to a moving platform by six prismatic legs with spherical joints. In the octahedral (3-3) form each of three platform nodes is joined to two base joints in a zigzag pattern (Figure 4); its forward kinematics reduces to an octic polynomial [4], and distance-based formulations avoid trigonometric substitutions [5]. For contrast, the general 6-6 platform has up to 40 solutions [6]. In the example test instance used in this work, the fixed base spherical joints are $B_1 = (0, 0, 0)$, $B_2 = (6, 0, 0)$, $B_3 = (2, 5, 0)$ and the moving nodes are $P_i = (x_i, y_i, z_i)$, $i = 1, 2, 3$, connected as in Table 4. The goal is to determine $P_1, P_2, P_3 \in \mathbb{R}^3$: nine equations in nine unknowns.

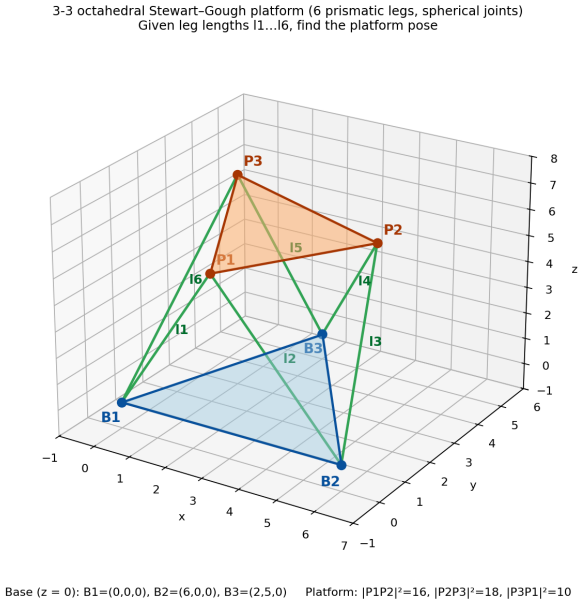


Figure 4: Octahedral (3-3) Stewart–Gough platform. Leg labels $l_1, \ldots, l_6$ follow Table 4.

Leg	l_1	l_2	l_3	l_4	l_5	l_6
Connects	P_1–B_1	P_1–B_2	P_2–B_2	P_2–B_3	P_3–B_3	P_3–B_1
Squared length	21	45	$\frac{249}{5}$	18	$\frac{648}{13}$	$\frac{3801}{65}$

Table 4: Leg connectivity and squared leg lengths; platform edges satisfy $\|P_1 - P_2\|^2 = 16$, $\|P_2 - P_3\|^2 = 18$, $\|P_3 - P_1\|^2 = 10$.

5.2 Kinematic formulation to polynomial system

Each leg gives a sphere condition $\|P_i - B_j\|^2 = l_k^2$, and the rigid platform gives three edge conditions. Expanding, e.g. l_2: $(x_1 - 6)^2 + y_1^2 + z_1^2 = 45 \Rightarrow x_1^2 - 12x_1 + y_1^2 + z_1^2 - 9 = 0$; l_3: $(x_2 - 6)^2 + y_2^2 + z_2^2 = \frac{249}{5} \Rightarrow x_2^2 - 12x_2 + y_2^2 + z_2^2 - \frac{69}{5} = 0$; l_4: $(x_2 - 2)^2 + (y_2 - 5)^2 + z_2^2 = 18 \Rightarrow x_2^2 - 4x_2 + y_2^2 - 10y_2 + z_2^2 + 11 = 0$; l_5 gives the constant $-\frac{271}{13}$. With variable ordering $x_1 > y_1 > z_1 > x_2 > y_2 > z_2 > x_3 > y_3 > z_3$ the system is

$$f_1 = x_1^2 + y_1^2 + z_1^2 - 21, \qquad f_2 = x_1^2 - 12x_1 + y_1^2 + z_1^2 - 9,$$
$$f_3 = x_2^2 - 12x_2 + y_2^2 + z_2^2 - \tfrac{69}{5}, \qquad f_4 = x_2^2 - 4x_2 + y_2^2 - 10y_2 + z_2^2 + 11,$$
$$f_5 = x_3^2 - 4x_3 + y_3^2 - 10y_3 + z_3^2 - \tfrac{271}{13}, \qquad f_6 = x_3^2 + y_3^2 + z_3^2 - \tfrac{3801}{65}$$

$$f_7 = (x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2 - 16,$$
$$f_8 = (x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2 - 18,$$
$$f_9 = (x_3 - x_1)^2 + (y_3 - y_1)^2 + (z_3 - z_1)^2 - 10.$$

5.3 Algebraic solution

The `lex` Gröbner basis has nine elements, with a triangular structure. The first is $x_1 - 1$ (indeed $f_1 - f_2 = 12x_1 - 12$), so $x_1 = 1$ for all solutions. The next seven express $y_1, z_1, x_2, y_2, z_2, x_3, y_3$ as polynomials in z_3 with rational coefficients of large height (omitted), and the last is the univariate polynomial in z_3 of degree 16, which factors as

$$(169 z_3^2 - 7744)\, h(z_3^2) = 0, \qquad \deg h = 7 \text{ in } z_3^2.$$

It is an octic in $u = z_3^2$, so the 16 complex solutions come in mirror pairs $z \mapsto -z$ (reflection through the base plane) [4]. The octic has exactly two positive real roots, $u = \frac{7744}{169}$ and $u \approx 46.2056$; the others are negative or non-real and give non-real z_3. Hence $z_3 = \pm\frac{88}{13}$ and $z_3 = \pm 6.79747$, and back-substitution gives four real solutions, two mirror pairs (Table 5). For solutions 1 and 2 the coordinates are exact: $P_1 = (1, 2, \pm 4)$, $P_2 = (\frac{17}{5}, \frac{26}{5}, \pm 4)$, $P_3 = (\frac{44}{65}, \frac{227}{65}, \pm\frac{88}{13})$. The maximum equation residual of each listed solution is below 1.1×10^{-14}.

No.	$P_1 = (x_1, y_1, z_1)$	$P_2 = (x_2, y_2, z_2)$	$P_3 = (x_3, y_3, z_3)$
1	(1, 2, 4)	(3.4, 5.2, 4)	(0.67692, 3.49231, 6.76923)
2	(1, 2, −4)	(3.4, 5.2, −4)	(0.67692, 3.49231, −6.76923)
3	(1, 2.40432, 3.77084)	(3.64896, 5.39917, 3.88865)	(1.53211, 3.15023, 6.79747)
4	(1, 2.40432, −3.77084)	(3.64896, 5.39917, −3.88865)	(1.53211, 3.15023, −6.79747)

Table 5: Real solutions of the octahedral platform (four of sixteen complex solutions); solutions 2 and 4 are the mirror images of 1 and 3 under $z \mapsto -z$.

All four are physical in the sense that the platform lies in $\mathbb{R}^3$; solutions 1 and 3 have the platform above the base plane ($z > 0$).

5.4 Computational benchmarks

Ordering	Macaulay2 (ms)	Python (s)	Python / Macaulay2
`lex`	34.7	more than 600 (10 min)	$> 1.7 \times 10^4$
`grlex`	16.7	95.099	$\approx 5.7 \times 10^3$
`grevlex`	9.62	92.248	$\approx 9.6 \times 10^3$

Table 6: Execution time for the Stewart–Gough system. The Python `lex` run did not finish within 10 minutes.

This is the hardest of the three systems. The Python `lex` run did not finish in more than 600 seconds, whereas Macaulay2 needed 34.7 ms. In Macaulay2, `grevlex` is the fastest ordering ($\approx 1.7\times$ faster than `grlex` and $\approx 3.6\times$ faster than `lex`); in Python the two graded orderings are within about 3% of each other.

6 Conclusion

Every system in this report was reduced to a polynomial system $f_1, \ldots, f_n$ and solved through a `lex` Gröbner basis. The 2R1P instance has two physical inverse-kinematics

solutions; the 3-RPR instance has one real pose $(x, y, \phi) = (0, 1, 0°)$ among six complex roots counted with multiplicity; and the octahedral platform has four real poses (two mirror pairs) among sixteen complex ones. In all three, the `lex` basis gives a triangular system from which solutions follow by back-substitution.

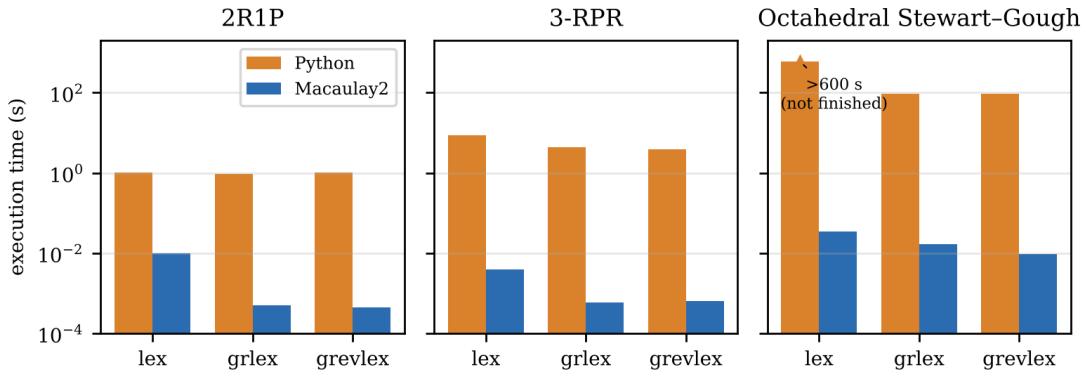


Figure 5: Execution time (logarithmic scale) by monomial ordering for each mechanism.

Figure 5 summarises the timings. Macaulay2 is faster than the Python implementation by about two to four orders of magnitude, and `lex` is the most expensive ordering in Macaulay2 for all three mechanisms; the Python `lex` run on the Stewart–Gough system was abandoned after more than 600 seconds. Python times are whole-script run times of an instrumented implementation, so the ratios should be read as practical rather than purely algorithmic. Natural extensions are the general 6-6 Stewart–Gough platform with its 40 solutions [6], parametric families of 3-RPR manipulators [3], and the conversion route from `grevlex` to `lex` for larger systems.

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